Machine Learning

Problem Set 04

Question 1. Linear, Ridge and LASSO

Let $N = 2$, $X = \{x_1 = (2, 1), x_2 = (4, 0)\}$, $Y = \{y_1 = 1, y_2 = 3\}$, and $\lambda = 1$. Compute each minimum value and give the corresponding minimizer:

(a) $\min_w \frac{1}{2N} \sum_{i=1}^{N} \left(w^\top x_i - y_i \right)^2$

(b) $\min_w \frac{1}{2N} \sum_{i=1}^{N} \left(w^\top x_i - y_i \right)^2 + \frac{\lambda}{2} \|w\|_2^2$

(c) $\min_w \frac{1}{2N} \sum_{i=1}^{N} \left(w^\top x_i - y_i \right)^2 + \frac{\lambda}{2} \|w\|_1$

Question 2. Learning and the No-Free-Lunch Theorem

Recall the following statements from Lecture 5: Corollary 2.3 (slide 15), the definition of PAC learning (slide 22), and Theorem 6.8 (slide 36).

Corollary 2.3. Let H be a finite hypothesis class. Let $\delta \in (0, 1)$ and $\epsilon > 0$, and let m be an integer satisfying

$$m \geq \frac{\log(|H|/\delta)}{\epsilon}.$$

Then, for any labeling function f and any distribution D for which the realizability assumption holds (that is, for some $h \in H$, $L_{D,f}(h) = 0$), with probability at least $1 - \delta$ over the choice of an i.i.d. sample S of size m, every ERM hypothesis h_S satisfies

$$L_{D,f}(h_S) \leq \epsilon.$$

Definition (PAC learning). A hypothesis class H is PAC learnable if there exist a function $m_H : (0, 1)^2 \to \mathbb{N}$ and a learning algorithm with the following property: for every $\epsilon, \delta \in (0, 1)$, every distribution D over X, and every labeling function $f : X \to \{0, 1\}$, if the realizability assumption holds with respect to H, D, f, then running the algorithm on $m \geq m_H(\epsilon, \delta)$ i.i.d. examples generated by D and labeled by f returns a hypothesis h such that, with probability at least $1 - \delta$ over the choice of the examples,

$$L_{D,f}(h) \leq \epsilon.$$

The algorithm is not necessarily ERM. A finite hypothesis class H is PAC learnable by ERM.

Theorem 6.8 (The fundamental theorem of statistical learning—quantitative version). Let H be a hypothesis class of functions from a domain X to $\{0,1\}$, and let the loss function be the 0–1 loss. Assume that $\text{VCdim}(H) = d < \infty$. Then there are absolute constants C_1, C_2 such that:

- H is agnostic PAC learnable with sample complexity

$$C_1 \frac{d + \log(1/\delta)}{\epsilon^2} \leq m_H(\epsilon, \delta) \leq C_2 \frac{d + \log(1/\delta)}{\epsilon^2}.$$

- H is PAC learnable with sample complexity

$$C_1 \frac{d + \log(1/\delta)}{\epsilon} \leq m_H(\epsilon, \delta) \leq C_2 \frac{d \log(1/\epsilon) + \log(1/\delta)}{\epsilon}.$$

Answer the following questions.

(a) All three give positive conclusions about learnability. How do these learning guarantees get around the No-Free-Lunch barrier? What common principle underlies their approaches?

(b) Looking beyond the machine learning framework, is the principle identified in part (a) the only way to address the challenge raised by the No-Free-Lunch theorem? Could another perspective explain how powerful algorithms, such as today's large language models, can perform well across a wide range of real-world tasks despite the No-Free-Lunch barrier? Explain your reasoning in no more than **300 words**.